Jasco Circular Dichroism Spectrophotometer

- Install Date: 1/26/2022
  - quote
  - packing slip
- Model: J-1500-150
- Serial Number: D067961638
- Location: CB-016A
- Installer contact information:
- Instrument install CD (Windows 10): Jasco-CD-Installer.iso
  - Hardware Manual from installer CD
- Software Product Key: PLT-YMX-01301483
- Supplemental Peltier manual CD: isoFile
- Standard Operating Procedure: Not available yet

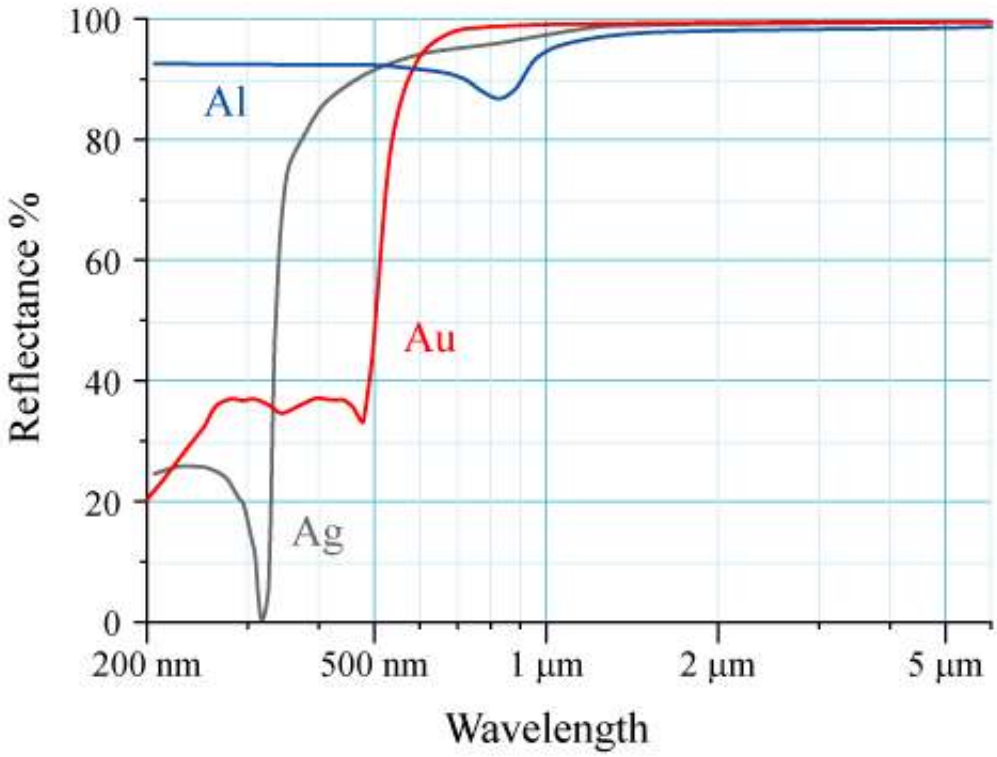
Nitrogen Generator

- Manufacturer: Apex
- Model: Nevis 10C
- serial number: 10283
- manufacturer contact: karen@apexgasgenerators.com
- Nitrogen Generator: manual

My Notes

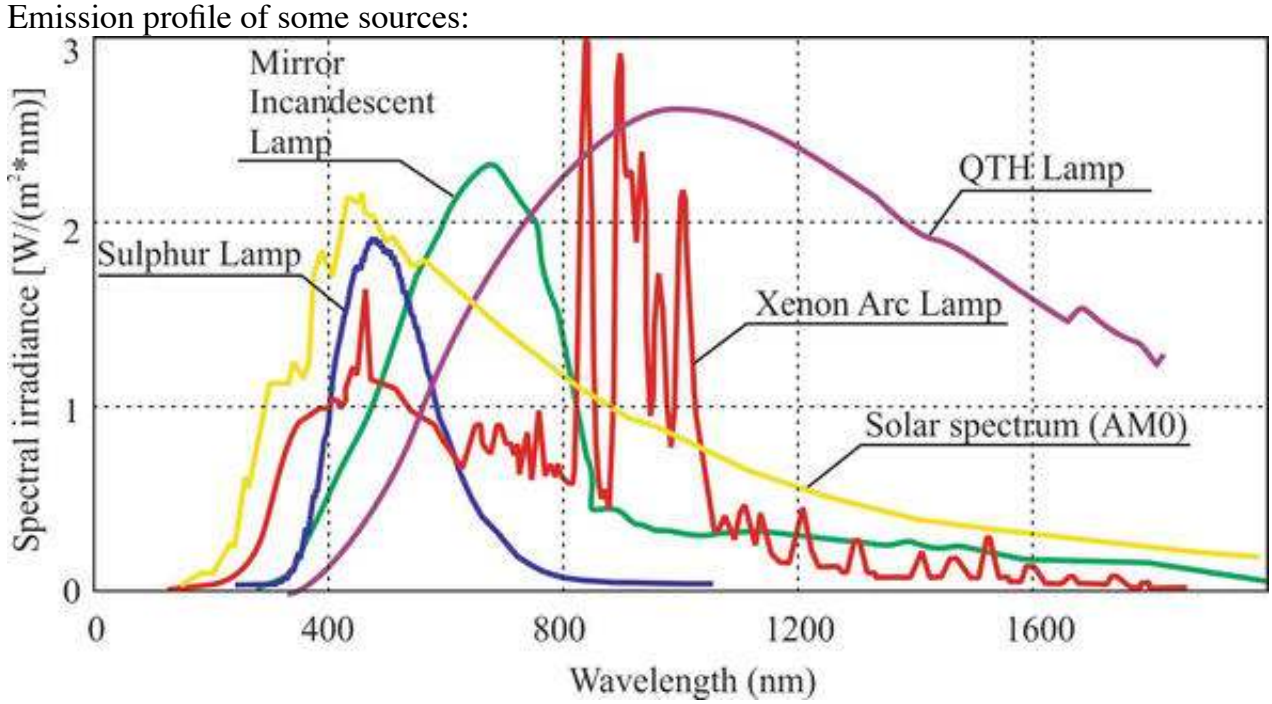
Nitrogen generator requires a 20 A circuit.

The reason they use aluminum mirrors. Aluminum has the highest reflectance in the UV/VIS region.



The reason you need a nitrogen purge

- In the UV below 240 nm the Xenon lamp can generare ozone from oxygen. Ozone can damage the optics.
- Oxygen abosorbs in the UV region. The xenon lamp already has very low emission intensity in that region
- Emission profile of some sources:



Linearly polarized light can be thought of as the superposition of right-hand circularly polarized light and left-hand circularly polarized light. See this youtube video for an excellent discussion.

How to disable automatic updates on Windows 10. I did this on the computer during the instrument install.

- launch services app
- find Windows Update
- select disable
- Click apply, ok, and stop

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Easton, MD 21601
W www.jascoinc.com
P 800 333 5272
F 410 822 7526
E sales@jascoinc.com
Please email orders to address above

QUOTATION

University of Nevada - Reno
Yftah Talgan
1664 N. Virignia St
Reno, NV 89557
775-784-4040

Warranty:	1 year parts and labor
Terms:	TBD
Shipping:	FOB, Origin, charge back buyer
Delivery:	TBD

Quote Number	Quote Date	Quote Validity	Sales Representative
Q-31271-H3J6-3	6/21/2021	6/21/2021 to 7/21/2021	Wayne Kottkamp

Part Number	Description	Unit Price	Qty	Extended Price
7000-J005A	**J-1500 Circular Dichroism Spectropolarimeter** **Specifications** • Optics: Dual prism monochromator with collimated beam • Measurement modes: CD, LD and ABS (many optional modes available) • Wavelength range: 163 - 950nm (up to 1600nm optional) • Stray light: Less that 0.0003% at 200nm • RMS Noise: Less than 0.004mdeg at 185nm • Light source: Xe 150W, supracil, air-cooled • Scan modes: Continuous scan, step-scan and auto-scan • Integrated wavelength calibration source with optimized N2 purge • Automated accessory recognition system • USB2 communication port **Contains** • J-1500 Main Unit • Complete 64 bit Spectra Manager II software suite provided (Spectra Manager CFR optional) • 10 and 1mm cells • N2 flowmeter • Tool kit • On-site warranty including parts, labor and travel • Complimentary attendance at two day advanced CD applications seminar • **DOES NOT INCLUDE PC REQUIRED FOR OPERATION**, unless quoted below	$82,500.00	1	$82,500.00
7069-J011A	**MPTC-511 six position in-line Peltier temperature control system** **Specifications** • Temperature range: -40 to +130˚C • Temperature accuracy: +/- 0.5˚C • Designed for use with up to six 10, 5, 2 and 1mm cells **Contains** • MPTC-511 Main Unit • Temperature probes for in-cell as well as adjacent to cell	$9,500.00	1	$9,500.00

Nevis 10C 110v
Nitrogen Generator
User Manual

Apex Gasgen Ltd
Elderpark Street
Glasgow G51 3TR
Scotland, UK

www.apexgasgenerators.co.uk

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Manual Revision		
Release Date	**Version**	**Record of Revision**
20/07/18	*2C*	*Forth Release*

This manual contains technical information to enable operation, maintenance, troubleshooting and repair of the system.

In all communications with Apex please quote the serial number of the system you are referring to. The serial number can be found on the CE/Services label attached to the rear of the enclosure.

For easy reference, copy the serial numbers into the boxes below:

Generator Model Type:	
Serial Number:	

Disclaimer

Apex Gasgen Ltd has written this manual to be an easy to use guide for the Nevis nitrogen generator range. All statements, technical information, recommendations in this manual and in any associated guides or related documents are written in good faith. However the accuracy and the completeness thereof are not guaranteed or warranted. They are not intended to be, nor should they be understood to be, representations or warranties concerning the products described. This nitrogen gas generator has been sold subject to the limited warranties set forth in the warranty statement. Further, Apex reserves the right to make changes in the specifications of the products described in this manual at any time without notice and without obligation to notify any person of such changes. This laboratory gas generator is designed for operation only by trained personnel familiar with the use of similar equipment and with safety requirements for the use of nitrogen and other industrial gases. Before operating this equipment, make sure to read and understand this information provided herein. If you have any questions, please contact Apex Gasgen Ltd or your generator supplier.

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1. Introduction

Warnings

1. When required to open the enclosure for repair or maintenance, ensure that the unit is disconnected from the electrical mains supply.

2. No unqualified or untrained personnel should operate, repair or maintain the system.

3. Apex shall not be liable for damages to the customer's own equipment which is the result of the contravention of the products specifications, warnings, cautions and notes, and the common sense of good engineering practice.

4. Do not disconnect or exchange any pipes or cables while the system is powered or under pressure.

5. It is the user's responsibility to observe all safety rules and observe the system product specifications.

6. Do not insert hand or other objects inside the machine unless it is fully isolated from the mains power supply. There are multiple cooling fans located near to the compressor.

2. Installation

Material Listing

After unpacking your Generator, you should have the following components:

- ✓ Nevis Nitrogen Generator
- ✓ Cabinet enclosure key (1 off)

Positioning the Generator

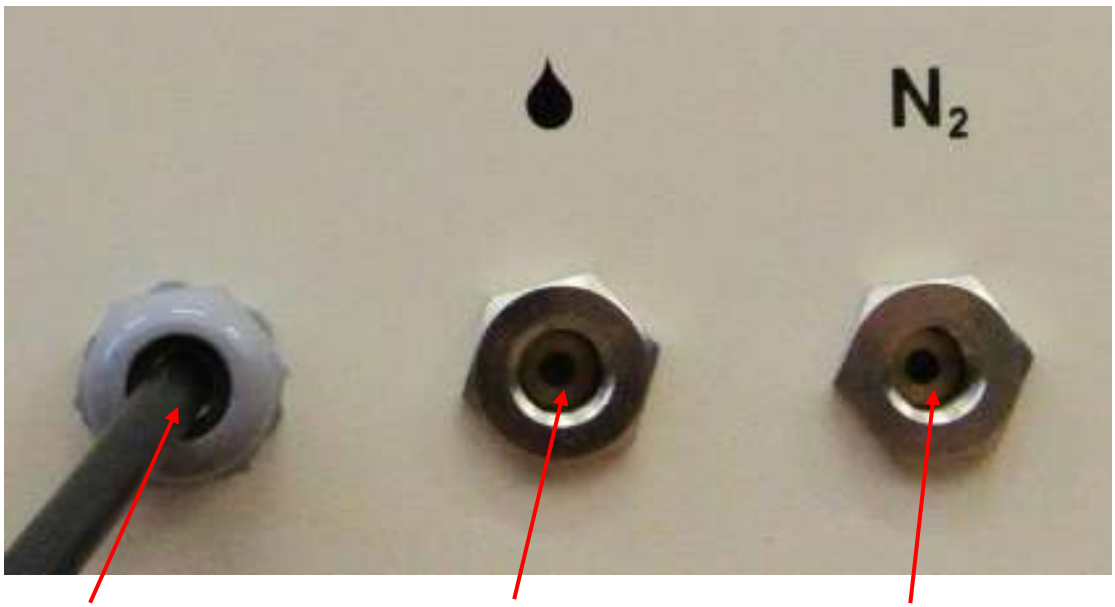
Your Nevis Nitrogen Generator has been designed to conform to most industrial and laboratory environments. The height has been selected such that the system will, in many cases, fit neatly below standard workbenches.

The Nitrogen Generator should be positoned in a convenient location, reasonably close to the application and near an appropriate electrical socket. Alternatively, in view of the fact that the compressor within the unit will start and stop regularly some users may prefer to install the unit in a separate “Machine” room and pipe the outlets to the application. Some variants of the machine are fitted with a compressed air inlet such that the machine can be operated independently of the inbuilt air compressor if required.

We would always recommend positioning the generator such that airflow through the door vents is not obstructed. The air compressor within the unit will generate heat during operation and it is essential that the ventilation through the machine remains unobstructed. The machine has been designed such that the cooling air required will flow from left to right when viewed from the front of the machine.

Connecting the Generator

The connection panel is to be found at the rear of the machine and will appear as follows:



1. Electrical Power Connector
Connect this cable to the mains supply which should be 110-120V/60HZ AC. The supply should be rated to **20 amps**.

2. Moisture outlet Connection
This drain should be connected with suitable piping, the thread is 1/4" BSPP female. Moisture removed from the compressed air is ejected here and should be drained to a suitable receptacle. (Not supplied)

3. Outlet Nitrogen Connection
This should be connected to your application with suitable piping, the thread is 1/4" BSPP female.

Basics – The Generator

Front:

Main Control Panel
The master control panel for controlling the operation of the compressor and generation of nitrogen is located behind the door to this side of the generator

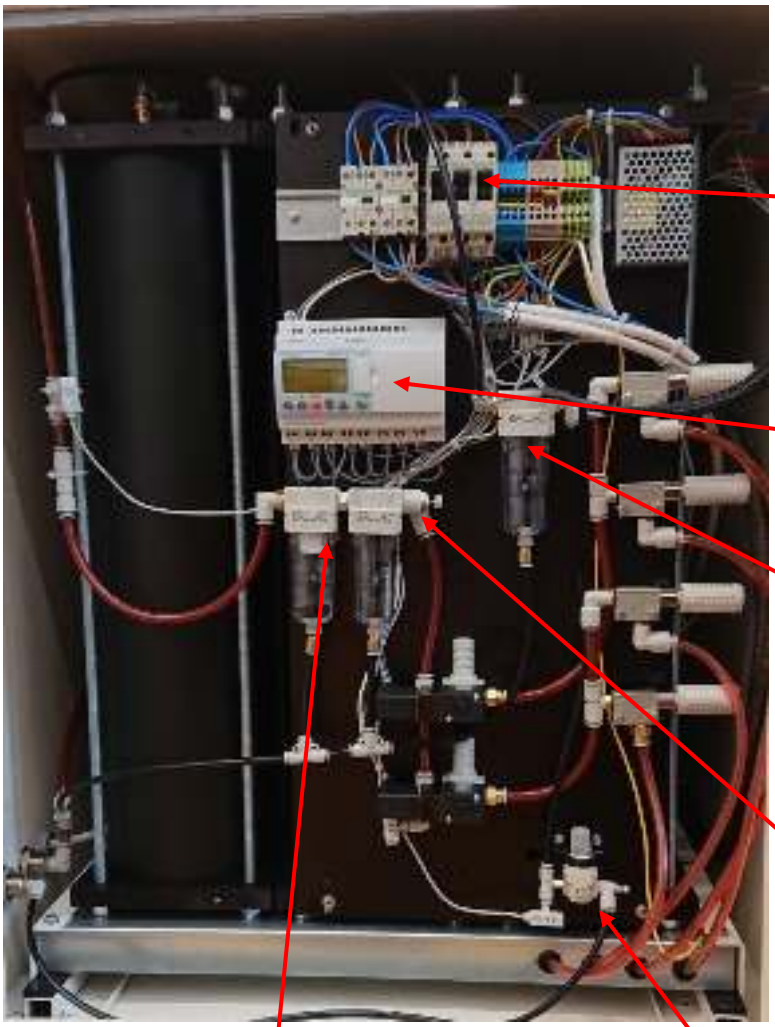
Compressor
The compressor is located behind the door panel to this side of the generator.

On Off Switch
This switch is used to switch the generator on or off. When set to position 1, the switch will light and the generator will be operational. Similarly position 0 switches the generator off completely although this does not isolate the machine from the mains supply.

As described above, the control panel for the system is behind the removable door fitted to the left hand side of the generator. This door can be removed by unlocking (turn clockwise) the two locks using the security key supplied and can then be lifted out completely to gain access to the control panel. To refit the panel, place bottom edge of the door into the door access way and close panel. Turn both locks anticlockwise to fix the door panel in place.

WARNING – There is a cooling fan within the generator and care must be taken not to insert hands of other objects beyond the control panels unless the machine has been fully isolated from the mains supply and the fan has stopped running. This is particularly important when accessing the right hand side of the unit which houses the compressor unit.

Control Panel – Key Components



Circuit Breakers controlling the compressors (Left) and 24 volt systems and 230 volt fans (Right).

PLC Controller controlling the system.

0.1 Micron air filter to remove any particles post nitrogen generation and prior to external equipment.

Regulator and Flow Controller controlling the pressure and flow of compressed air into the nitrogen generator.

0.1 & 1.0 Micron air pre-filters to remove any particles prior to nitrogen generation.

Flow controller and regulator controlling flow and pressure of nitrogen leaving the generator.

Note:- The regulator and the air in flow filter are factory set to maximise the efficient operation of the nitrogen generator and should require no further adjustment. The nitrogen outlet regulator and flow controller is preset at the factory to allow a maximum of 10l/pm at 80psi/5.5bar.

The control panel is located at the left hand end of the machine when viewed from the frontal perspective. The key elements of this control panel are as outlined below:

Circuit Breakers – The circuit breakers control the key elements of the generator as follows:

No1. Left hand circuit breaker controls the power to the left hand compressor which will switch on and off as directed by the PLC controller.

No2. Middle circuit breaker controls the power to the right hand compressor which will switch on and off as directed by the PLC controller.

No3. Right hand circuit breaker controls the power to the 24 volt circuits within the system and the 230 volt supply to the cooling fans.

To power up the system, first ensure that the lighted switch on the front of the generator is in the on or “1” position and the switch is illuminated.

To start the generator: Go to No3 circuit breaker/ right hand circuit breaker and switch to the on position by pushing the switch upwards. Check that the green LED on the outside of the generator is now illuminated. Now go to No1 circuit breaker/left hand circuit breaker and switch to the on position by pushing the switch upwards. The left hand compressor will now start. Finally go to the No2 circuit breaker/middle circuit breaker breaker and switch to the on position by pushing the switch upwards. The right hand compressor will now start.

PLC Controller - This controls the operation of the entire system and can be used to display information on how the system is functioning. The display panel can be illuminated for a short period of time by pushing the green button briefly. The “A” and “B” buttons can be used to cycle through the Seven display screens either forward using the “A” button or in reverse using the “B” button.

Inlet and outlet Air particle filters – The elemnts in these filters require to be replaced on an annual basis (more frequently if the generator is in a dirty environment) and replacement filter elements can be supplied by your Apex distributor or by contacting Apex directly.

3. Operation

Operation Summary

The Nevis Nitrogen Generator is fitted with an energy-saving system that only produces Nitrogen when you require it, while at the same time ensuring that when you do require Nitrogen it is of a consistent purity. This means that you should not switch the Nevis Nitrogen Generator off as it periodically purges the system to maintain purity unless you do not require nitrogen for a prolonged period of time.

The Nevis Nitrogen Generator is fitted with an integral air compressor, and when the unit is not required to produce Nitrogen, the compressor will only switch on when called upon to supply the CMS columns with air. When there is no call for nitrogen production, both generator and compressor will go into "standby" mode.

Generator lights

Your Nevis Nitrogen Generator has several lights that illuminate to provide information about your system

Power Switch - The power to the generator is switched on using the rocker switch on the front of the machine . When this is switched to the left, the green light inside the switch will glow red and the machine will be powered up.

Green Power LED – This light comes on indicate that 24VDC is being supplied after circuit breaker No2/right hand circuit breaker is switched on.

Blue N2 LED – This light comes on when the Nitrogen within the system has reached the specified delivery pressure and stays on till the generator enters the standby mode or is powered down or drops below the specified delivery pressure.

Red Alarm LED- This illuminates when the compressor running time exceeds 2000hrs and indicates that the system requires a service. This light can only be turned off by Apex, Apex certified service personnel or under their instruction.

Start Up

Status indication throughout the different phases of the operation of the Nitrogen Generator

Purging

Initially the Nevis Nitrogen Generator does not have any nitrogen stored and its nitrogen reservoir is full of air. As the Nevis Nitrogen Generator starts to add nitrogen to its reservoir the nitrogen content will start to increase.

During the purging period, the N2 Outlet Port on the rear of the generator should be left open to atmosphere when it is running to allow the unit to purge itself to the desired outlet purity. Below is a table of the approximate purge times for Nevis Nitrogen Generator.

Nevis Nitrogen Generator System Purity	Purge Time
99.9995%	5 hours

After the initial purging period, you should connect your application to the N2 Outlet Port permanently.

Shut down

The electrical supply must not be cut off from the generator suddenly. The procedure for shut down is to close off the outlet flow of nitrogen from the generator; this will result in the generator going into standby operation and if the compressor has been in operation, it will stop automatically. Standby is indicated by the blue N2 LED on the front of the generator no longer being illuminated.

The rotary switch should only be switched to the 0 or “Off” position when the generator is not going to be used for some considerable time. It is worth bearing in mind that if the generator is switched off for a prolonged period of time you will be required to go through the purge process detailed above in order to ensure the level of purity of nitrogen is appropriate for your requirements.

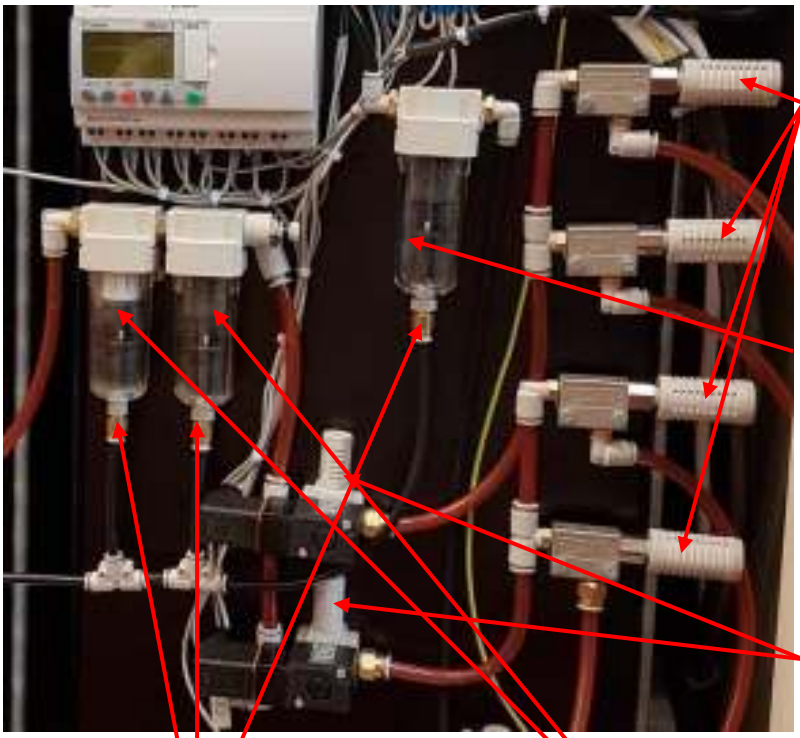
4. Maintenance

Servicing – Nitrogen Generation Parts

Filters and Silencers

The filter elements within all three filters and the system silencers should be renewed once a year or when the Alarm light illuminates. If the system operates in a dusty or dirty environment we would recommend that the filters and silencers are changed on a more frequent basis without waiting for the Alarm light to come on.

Ensure that your air supply and the power supply are switched off before any maintenance work is carried out. Any pressure left in the line can be drained by turning the auto drain anticlockwise to the manual override position. Please ensure that the auto drain is reset to automatic again prior to starting up the machine otherwise compressed air will run continuously through the system drain.



Large Silencers
Replace the four silencers. Refer to page 17 for part numbers.

Outlet Air Filter
Replace the black air filter element on an annual basis or more often if required. Refer to page 17 for part numbers.

Small Silencers
Replace the two silencers. Refer to page 17 for part numbers.

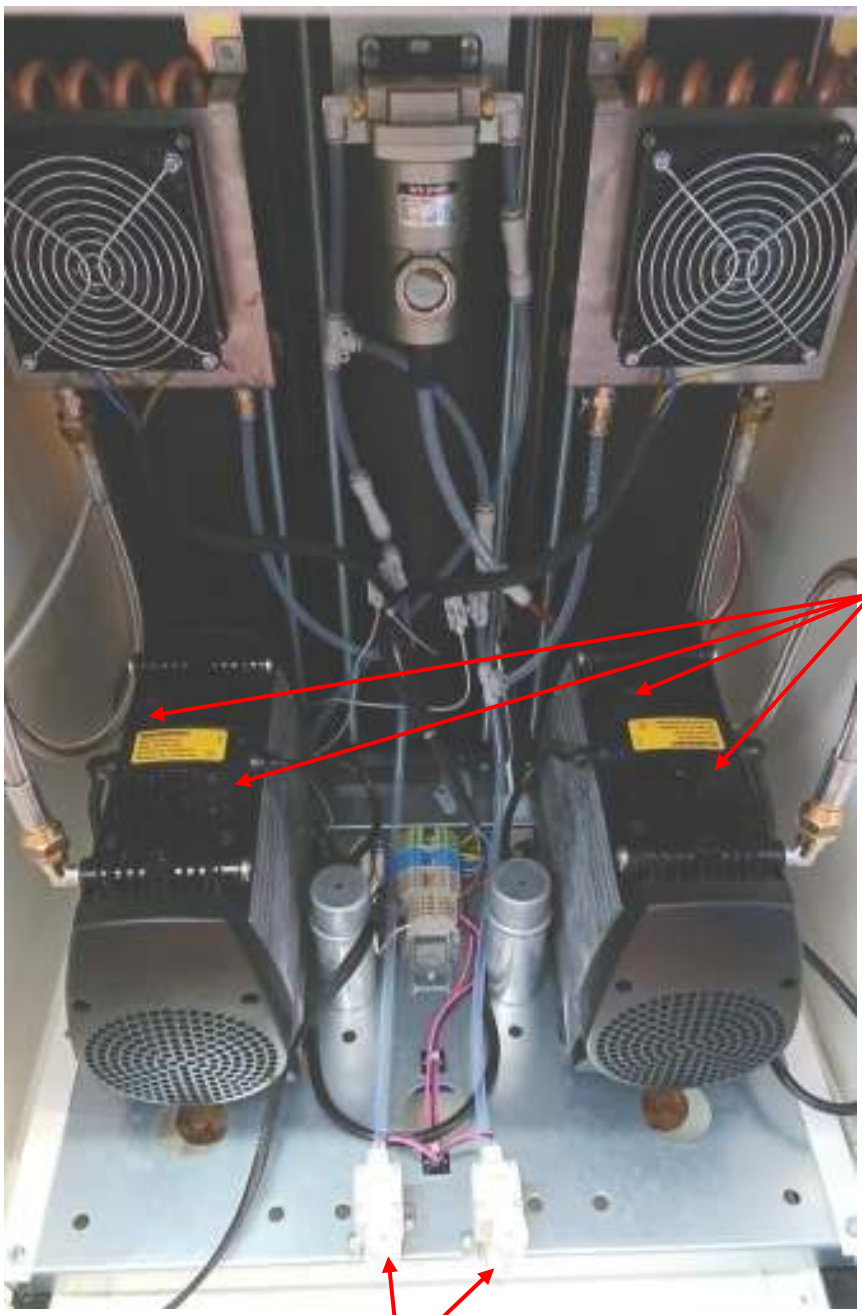
Auto Drain – Manual override
Turn anticlockwise to engage manual override. Remember to turn clockwise to engage the auto drain once you have finished maintenance.

Inlet Air Filters
Replace the white (right hand unit) and black (left hand unit) air filters on an annual basis or more often if required. Refer to page 17 for part numbers.

Servicing – Compressor Parts

To access the compressor part of the system you will need to open the Right hand side door of the system. You should note that there is a fan attached to this door and thus it is heavier than the other door. There should be enough free length on the electrical connection of the fan for the door to be safely put to the side of the system but if required the fan can be disconnected. This is done by first ensuring the system is switched off and ideally disconnected from its power supply and then undoing three small screw connectors for the fan's Live (+ve), neutral (-ve) and earth connections as below.

Fan power connections
Disconnect the 3 wires from the fan, Brown is +ve, Blue is –ve and yellow/green is neutral. Do not forget to reconnect these once you have completed the service.



Compressor – Air Filters
Remove and replace the intake filters located as indicated. Refer to page 17 for part numbers. These Are obtained from Apex and should be replaced on an annual basis or more frequently if required.

Compressor – Process valve Silencers
Replace the large silencer. Refer to page 17 for part number.
This is obtained from Apex and should be replaced on an annual basis or more frequently if required.

5. System Control Interface

System PLC Controller

The PLC controller that controls the operation of the system is located on the front panel. This controller has a number of buttons of which only the following are operational:

1. Green Button – pushing this briefly has the effect of illuminating the panel for a brief period of time.

2. Button "A" – pushing this button briefly allows you to cycle forwards through the display screens that are available to the user.

3. Button "B" – pushing this button briefly allows you to cycle in reverse through the display screens that are available to the user.

The following screen prints detail the information available to the user:

Welcome screen detailing the version of the machine together with software release currently installed.

This displays the current pressure within the operational CMS columns and the nitrogen receiver (RES). The display provides readings measured in both Barg and PSI.

This displays the current air pressure in the compressor air reservoir, measured in both Barg and PSI.

This displays the current air pressure being produced by both compressors, measured in both Barg and PSI.

Total hours and minutes of nitrogen production since the unit was installed.

This screen details the specification of the air compressor installed within the unit.

Total hours and minutes of operation for the compressor since they were installed.

The serial number of the system

6. Service Parts

Generator Spare Parts (Excluding Compressor)

Apex Part Number		Part Description	Qty	Replacement Period
SP0200		**Service Pack for Nevis 10C Generator**	**1**	**Annually**
203456	Included	Black filter element for inlet and outlet filter	2	
203457	Included	White filter element for inlet filter	1	
202086	Included	Silencer 1/8" Large	6	
100007	Included	Silencer 1/8" Small	2	

Compressor Spare Parts

Apex Part Number		Part Description	Qty	Replacement Period
SP0201		**Intermediate Service Kit for Jun Air Compressor.**	**1**	**Annually**
300043	Included	Compressor air intake filter	4	
SP0205		**Comprehensive Servcie kit for Jun Air Compressor**	**1**	**After 4 years**
300156	Included	Cylinder head gaskets for compressor	2	
300162	Included	Compressor Con Rod assembly	2	

7. Appendices

Troubleshooting

Appendix A Flow Measurement/Adjustment

Connect a flow meter (to the Nitrogen outlet port via a length of tube (we usually use 6mm polyurethane) Make sure the needle valve on the flow meter is fully open. Leave the outlet of the flow meter to atmosphere, do not restrict.

Use a 0-30 l/min or similar flow meter for accuracy.

Adjust the nitrogen outlet flow controller on the nitrogen regulator to change flow and set your systems output. Remember this generator is designed for 10l/pm if you draw more nitrogen than that the purity will decline.

Note: if no flow when adjusting, the pressure regulator may be fully closed

Appendix B Pressure Measurement/Adjustment

Connect a pressure gauge to Nitrogen outlet (0-10 bar pressure gauge)

Outlet pressure regulator can be adjusted clockwise to increase pressure, counter clockwise to decrease pressure. To make any changes pull the top handle of the regulator upwards to unlock it so that adjustments can be made.

Note: if no pressure when adjusting, the outlet flow controller may be fully closed

Appendix C Inlet Flow Control Adjustment

Flow control should be fully open.
When adjusting inlet flow control, always take note of the factory preset position, by counting the amount of turns closed or opened when adjusting. The purpose of this flow control is to restrict the inlet air flow as much as possible while still reaching the required pressure by the end of the ON cycle. This can be seen on the LCD display under the reservoir(RES) pressure reading, the pressure should reach 105 psi within the last 20 seconds of the ON cycle (before the pressure is exhausted).
To set this, if the maximum pressure reading on the LCD reservoir pressure is 105 psi or more. Record that pressure, and then close the inlet flow controller until this pressure reduces. (adjust by only 1/8 of a turn, then let the generator run for a few cycles before you can see the effect). Once the flow control is restricting the inlet flow enough, take the flow control back one increment.

LCD Display
On the control panel there is an electronic PLC which runs a program controlling the pneumatics on the control panel. This also displays the pressure readings from a pressure transducer on the inlet and outlet pipes. The pressure readings are displayed on the LCD display on the PLC:-

Cycle Explanation
Basis of the cycle is a timer cycle, which has an ON cycle 180 sec (valve opens to let compressed air into the CMS column). OFF cycle for 180 secs (valve closes stopping compressed air feed to the CMS column, and dumps all the pressure in the CMS column to atmosphere at the same time).
With 2 CMS columns the cycle alternates, so that as one CMS column fills the other is venting.

Standby Operation
If no flow is taken from the nitrogen outlet therefore the reservoir (outlet pressure on LCD) does not drop, then the program will recognise this and go into standby mode, where the generator will shut down and only run 1 cycle per hour (to maintain purity). When the generator does not demand air from the compressor the compressor will also go into standby.

8. Warranty

LIMITED WARRANTY

LIMITED WARRANTY: APEX GASGEN LTD. ("**APEX**") warrants that the ITEMS LISTED BELOW shall be free from defects in material and workmanship for the stated period of time commencing from date of shipment or as stated below:

APEX GAS GENERATOR SYSTEM: One (1) year from customer receipt

Repair or replacement parts for Apex Gas Generator System: Ninety (90) days from customer receipt

EXCLUDED FROM THIS LIMITED WARRANTY:
The following shall be excluded from the Limited Warranty:

- Parts and items considered consumable in normal operations, including those parts and items supplied with the Apex Gas Generator System ("System") for maintenance.
- Any System and its parts that are not installed, operated, and maintained in accordance with the unit's manual supplied with the System.
- Damages due to accident, abuse, acts of God, acts of terrorism, misuse or negligence, or which result, in whole or in part, from improper or unauthorized use or repair of the System, or use of the System in a manner for which it was not designed, or by causes external to the System such as, but not limited to, power or air conditioning failure or voltage irregularities.

REMEDY: BUYER'S sole and exclusive remedy in the event of defect, and the liability of APEX hereunder is limited to the adjustment, repair, or replacement of the defective item or part with a similar item or part free of defect.

Such adjustments, repairs, or replacements will be made at APEX's Glasgow, Scotland plant or, by agreement of both parties, at the site of the System, if BUYER so elects. All costs for shipping equipment or parts shall be on the account of the BUYER whether to or from the point of manufacture. Labour costs associated with travel, expenses, and subsistence costs for field services shall be on the account of the BUYER.

VOIDING OF THE LIMITED WARRANTY: This Limited Warranty is immediately void upon:

- THE OPENING OR DISASSEMBLY OF THE SYSTEM CABINET (OR ANY PART THEREIN), OR
- THE SALE, ASSIGNMENT OR ANY OTHER TRANSFER OF TITLE BY BUYER OF THE ITEMS OR PARTS OTHERWISE COVERED UNDER THIS LIMITED WARRANTY

WAIVER OF ALL OTHER WARRANTIES: THE LIMITED WARRANTY PROVIDED HEREUNDER AND THE RIGHTS AND REMEDIES OF THE BUYER HEREUNDER ARE IN LIEU OF, AND BUYER EXPRESSLY WAIVES, ALL OTHER WARRANTIES, GUARANTEES, OBLIGATIONS, LIABILITIES, OR REMEDIES, EXPRESSED OR IMPLIED, ARISING BY LAW OR OTHERWISE, INCLUDING WITHOUT LIMITATION IMPLIED WARRANTIES OF MERCHANTABILITY AND NON-INFRINGEMENT, IMPLIED WARRANTIES ARISING FORM THE COURSE OF DEALING OR USAGE OF TRADE AND IMPLIED WARRANTIES OF SUITABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

LIMITATION OF LIABILITY: THE REMEDIES PROVIDED IN THIS LIMITED WARRANTY ARE EXCLUSIVE AND APEX SHALL IN NO WAY BE LIABLE FOR INCIDENTAL OR CONSEQUENTIAL DAMAGES OF ANY KIND WHATSOEVER INCLUDING WITHOUT LIMITATION LOSS OF USE, REVENUE OR PROFIT.

9. Service

Contacting Apex Gasgen Service Department

Apex provides worldwide engineering and technical field service support for all its products. Field Service representatives are available for contact at the addresses below:

Worldwide:

Apex Gasgen Ltd
Unit 36 Elderpark Workspaces
Elderpark Street
Glasgow
Scotland
G51 3TR
Tel: (44) 0141 530 7417

www.apexgasgenerators.co.uk
sales@apexgasgenerators.co.uk

	and internal for safety override • Built-in variable speed magnetic stirrer for all six positions • 1mm spacer • Temperature control software for temperature scan and temperature/wavelength scan programming • Instruction & Maintenance Manual			
EX2-1055 KOOLANCE	**Koolance Cooling Kit for use with water circulated peltier accessories** • Compression fittings, push-on mounting for 6-10 mm polymer tubing • Low profile, 2-Fan water circulation system for Peltier accessory heat-sink applications • Water stabilizer and anti-fungal agent • 110/220V AC power adapter • Black PVC 10 mm OD tubing for connection of Koolance circulator and Peltier accessories (6 meter length)	$610.00	1	$610.00
N10-C	**Nevis 10-C Nitrogen Generator** • Up to 10 L/min for rate • Up to 99.9995% purity • Built-in sound dampener • Built-in oil-free compressor • Simple low-cost maintenance	$13,155.00	1	$13,155.00
IT	**Installation & Training includes instrument and software familiarization** Performed by a factory trained JASCO Service Engineer Installation subject to satisfactory Completion of Site Requirements.	$3,175.00	1	$3,175.00
		Item Total:		$108,940.00
	Discount Type: Special	**Discount(%):**		15.00%
		Subtotal:		$92,599.00
	Notes:	**System Total:**		$92,599.00

Optional Products		Price per Unit
DELL OPTIPLEX 3080	**Dell OptiPlex 3080** • OptiPlex 30870 Desktop, Intel Core i5-8500 Processor • 8GB Ram, 2666 MHz • Wireless Keyboard and 2-Button Entry Mouse with scroll feature • Dell Professional P2219H 22" Monitor With LED • 500GB SATA HDD • DVD+/-RW Combo drive • Windows 10 Professional, 64-bit, English, Factory Install • ProSupport 3 Year Service Plan - Next Business Day Parts and Labor On-Site Response (provided by Dell); 24/7 Tech support	$1,405.00

For instruments purchased with installation by JASCO personnel training is provided. Additionally, JASCO offers regularly scheduled training seminars at our Easton, MD applications laboratory. Small classes and hands-on guidance ensure that you receive an optimum learning experience. Please visit www.jascoinc.com/training to view and register for available classes. On-site training at your facility is available upon request.

JASCO only quotes freight charges if specifically requested by the customer. Any freight charges included on this quote are estimated and are based on standard delivery – ground shipment. Actual freight charges invoiced may vary due to delivery method or carrier selected.

JASCO only quotes sales tax if specifically requested by the customer. Please refer to the attached Terms and Conditions, Prices and Quotations, item b for further information.

SALES TERMS AND CONDITIONS

DEFINITIONS
"Agent" or "Representative" - an employee or other person or entity authorized to action behalf of JASCO Incorporated (Company).
"Company" - JASCO Incorporated.
"Contract" - the official agreement formed between the Company and the Customer through the Company's acceptance of the Customer's order.
"Customer" - a person or entity entering into a contract with JASCO Incorporated for the purpose of purchasing goods or services from the Company.

FORMATION OF CONTRACT
a) No order in pursuance of any quotation shall be binding on the Company unless and until such order is accepted by the Company in writing. Orders will be within seven days from the date the verbal, telephone or fax submission was made to the Company.
b) The Contract shall be subject to the terms and conditions contained herein by reference. No Representative or Agent of the Company may bind the Company to any terms outside this agreement or make any representations inconsistent with such terms and conditions unless presented in writing and approved under delegated authority by the Company. A list of delegated authorities of the Company is available upon request.
c) Unless otherwise agreed in writing by the Company, these conditions shall override any terms and conditions stipulated by the Customer in reference to this order or other pre-contract negotiations.
d) Any description contained in the Company's catalogs, price lists or other promotional material is intended to present an accurate image of the Company's products or services. Catalogs, price lists or other promotional material, however, is not part of the Contract.
e) The Company reserves the right to correct any clerical or typographical errors included in the Contract at such time as the error is discovered by the Company. Revised contracts will be presented to the Customer, as necessary.
f) By accepting delivery of the goods supplied, Customer agrees to be bound by and accepts these terms and conditions unless Customer and Company enter into a separate written agreement signed by both parties. Any additional or different terms and conditions contained in any purchase order or other document provided by Customer are null and void. The terms and conditions contained herein encompass the entire understanding of the parties with respect to the subject matter hereof.

SAFETY INFORMATION AND USE
a) The Company warrants that it will communicate to all relevant personnel designated by the Customer all information pertaining to the safe use and handling of the goods included in this contract.
b) If the goods are used for purposes which have not been made known to the Company and approved by it in writing, are used in a manner other than is authorized by the Company, or are used in hazardous areas (unless approved by the Company), the performance and safety of the goods may be affected and the Customer, in such event, agrees to indemnify the Company against any claims, costs, damages or expenses arising from such unauthorized use, including (without prejudice to the generality of the foregoing) any infringement of any intellectual property rights arising there-from.

PRICES AND QUOTATIONS
a) Subject to any agreement to the contrary, the Company's quotations are valid for 30 days from the date of issue (errors and omissions excepted).
b) If, upon the Customer's request, goods and services provided under this quotation are revised, the Company shall amend the quotation to reflect the revised goods and services and related pricing thereof.
c) This quotation does not include freight charges, sales or other applicable tax, or duties or other government charges related to the delivery of goods and services to the Customer unless specifically requested by the Customer. However, Customer is responsible for these charges as part of the terms of this Contract and shall be invoiced for such by the Company and/or freight carrier upon the delivery of the goods and services to the Customer.
d) Where the Customer requires special packaging for the goods, the Company reserves the right to pass additional charges to the Customer without notice.

DELIVERY
The Company will deliver the goods by the means most convenient to the Company and to the address or addresses designated by the Customer. In the event that the Customer fails to specify a delivery address, the Company may deliver the goods to any address in North, South, or Central America at which the company resides or performs business activities.

RISK
a) Suitable unloading and storage of goods after delivery shall be the Customer's responsibility. Customer shall not open packages of products slated for Company installation. Improper unpacking of goods by the Customer may nullify the manufacturer's warranty replacement should any damage to the goods occur.
b) Subject to any agreement in writing by the Company, the risk in goods supplied by the Company shall pass to the Customer upon the date of shipment by the Company.

CONFIDENTIALITY
In the course of performing services, Company may be privy to confidential or proprietary information maintained, used, or created by Customer ("Confidential Information"). Company agrees to keep all Confidential Information in strict confidence. Company shall use the Confidential Information only to carry out its proper functions in connection with accomplishing the goals of the business relationship and shall not disclose the Confidential Information to any other individuals or entities. Confidential Information does not include information: (a) that was in the possession of Company without obligation of confidentiality prior to its receipt of Contract hereunder; (b) lawfully obtained by Company without restriction from a third party; or (c) publicly available other than through the fault or negligence of Company.

PAYMENT
a) Unless otherwise specified in writing by the Company, payment for goods or services or any installment thereof shall be made by the Customer at the net invoice price no later than 30 days after the invoice date.
b) Unless otherwise agreed in writing, the Company shall invoice the Customer for goods or services, or any installment thereof, upon shipment of the goods, upon notification to the Customer that the goods are ready for collection, or upon completion of the services, or any installment thereof.
c) Without prejudice to sub clauses (a) and (b) in this section, if the Contract provides for stage payments, the Company shall be entitled to payment at such intervals as specified in the Contract.
d) In the event the Company agrees to accept payment by installment, any delay of contractual payment by the Customer shall, without prejudice to any other right forthwith, immediately entitle the Company to the whole of the outstanding Contract sum.
e) Where the Company has agreed that (a) the price is payable in a custom other than US Dollars or (b) that payment will be in the form of a bank wire or other electronic payment, all banking fees and similar charges connected to the payment shall be borne by the Customer.
f) A late fee will be charged on all past due balances at the greater of $25 per month or 1 percent per month, beginning on the first day after the payment due date.
g) The Customer agrees to pay the Company legal fees and other costs incurred by the Company in connection with enforcement of the terms and conditions specified in this agreement.

DEFERMENT OR CANCELLATION
a) If the Customer fails to make any payment when it becomes due, or if there shall be any breach by the Customer of any of the terms and conditions of this Contract as referenced herein, the Company may defer or cancel any further deliveries of goods or services provided.
b) Cancellation of the Contract by the Customer must be in writing and will be accepted at the discretion of the Company.
c) Return of goods by the Customer requires return authorization issued by the Company. Unless the goods are damaged upon receipt (refer to DAMAGE, SHORTAGE OR LOSS below), the Company will charge a restocking fee of 25 percent for the return of goods by the Customer.
d) The Company shall charge the Customer for any costs incurred by the Company due to the suspension or deferment of any order by the Customer or in the event that the Customer defaults in collection any goods or giving instructions for the delivery of any goods or the performance of any service. The Company further reserves the right to invoice the Customer immediately in such events for the full price of the goods or services negotiated under the Contract.
e) No failure by the Company to enforce strict compliance by the Customer with such provisions shall constitute a waiver thereof and no termination of the Contract shall prejudice, limit, or extinguish the Company's rights under this paragraph.

DAMAGE, SHORTAGE OR LOSS
No claim for damage in transit, shortage of delivery, or loss of goods will be accepted unless the Customer provides written notice of such damage, shortage or loss (including reasonable particulars thereof) within 3 business days of receipt of the goods. For goods requiring installation by the Company, the Customer shall contact the Company within 3 business days of receipt to inform the Company of any damaged packages. Refer to the section entitled "Risk" contained herein for further information on unpacking goods. The Company may replace or repair the goods at its discretion and the Company's liability, if any, shall be limited to product damage, shortage or loss. Repair or replacement of goods shall be contingent upon the Customer's return of damaged goods to the Company within 14 days of such request by the Company. The Customer shall not be entitled to make any claim against the Company for consequential loss arising out of such damage, shortage or loss as aforesaid.

LIMITATION OF LIABILITY
a) UNDER NO CIRCUMSTANCES AND NOTWITHSTANDING THE FAILURE OF ESSENTIAL PURPOSE OR ANY REMEDY SET FORTH HEREIN, WILL COMPANY BE LIABLE FOR (A) ANY INCIDENTAL, INDIRECT, SPECIAL, PUNITIVE OR CONSEQUENTIAL DAMAGES INCLUDING, BUT NOT LIMITED TO, LOSS OF PROFITS, BUSINESS REVENUES OR SAVINGS, EVEN IF COMPANY HAS BEEN ADVISED OF THE POSSIBILITIES OF SUCH DAMAGES OR IF SUCH DAMAGES ARE OTHERWISE FORESEEABLE IN EACH CASE WHETHER A CLAIM FOR SUCH LIABILITY IS PREMISED UPON BREACH OF CONTRACT, WARRANTY, NEGLIGENCE, STRICT LIABILITY OR OTHER THEORY OF LIABILITY, (B) ANY CLAIMS DEMANDS OR ACTIONS AGAINST CUSTOMER BY ANY THIRD-PARTY, (C) ANY LOSS OR CLAIM ARISING OUT OF OR IN CONNECTION WITH THE CUSTOMER'S IMPLEMENTATION OF ANY CONCLUSIONS OR RECOMMENDATIONS BY COMPANY BASED ON, RESULTING FROM, ARISING OUT OF OR OTHERWISE RELATED TO THE PRODUCTS OR SERVICES, OR (D) ANY UNAVAILABILITY OF THE PRODUCT FOR USE OR ANY LOST, DAMAGED OR CORRUPTED DATA OR SOFTWARE.
b) Unless otherwise provided in these conditions, the Company's liability with respect to any defect or failure of goods supplied, or default in services provided is limited to repairing or paying for the repair or replacement of goods which within 12 months of dispatch are found to be defective by reason of faulty or incorrect parts or materials or default in any services which the Company has failed properly to perform in accordance with the Contract. **COMPANY EXPRESSLY DISCLAIMS ALL WARRANTIES EITHER EXPRESS OR IMPLIED, RELATED TO PRODUCTS, INCLUDING, BUT NOT LIMITED TO, ANY WARRANTY OF TITLE, ACCURACY, MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.** Customer acknowledges that no Employee, Agent or Representative of the Company is authorized to make any representation or warranty that is not contained in these terms and conditions on behalf of Company.
c) Conditions precedent to the Company's liability hereunder shall be that the Customer, as soon as reasonably practicable, shall provide the Company reasonable notice of the defect, failure, default or error and shall provide authority for the Company's employees or agents to inspect the goods. Company warrants that any services to be performed under the Contract will be performed in a good and workmanlike manner. Customer's sole and exclusive remedy and Company's entire liability with respect to this warranty will be, at the option of the Company to either (a) use its reasonable commercial efforts to re-perform or cause to be re-performed any service not in substantial compliance with this warranty or (b) refund amounts paid by Customer related to the portion of the service not in substantial compliance, provided, in either case, Customer notifies Company in writing within five business days after performance of the service. **EXCEPT AS SET FORTH HEREIN, AND SUBJECT TO APPLICABLE LAW, SELLER MAKES NO OTHER AND EXPRESSLY DISCLAIMS ALL REPRESENTATIONS, WARRANTIES, CONDITIONS OR COVENANT, EITHER EXPRESS OR IMPLIED (INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTY OF MERCHANTIBILITY OR FITNESS FOR A PARTICULAR PURPOSE) ARISING OUT OF OR RELATED TO THE PERFORMANCE OR NON-PERFORMANCE OF THE SERVICES, INCLUDING BUT NOT LIMITED TO ANY WARRANTY WITH RESPECT TO THE PERFORMANCE OF ANY HARDWARE OR SOFTWARE USED IN PERFOMING SERVICES AND ANY WARRANTY CONCERNING THE RESULTS TO BE OBTAINED FROM THE SERVICES. THIS DISCLAIMER AND EXCLUSION SHALL APPLY EVEN IF THE REMEDY SET FORTH HEREIN FAILS OF ITS ESSENTIAL PURPOSE.**

INSTALLATION
a) Where the Company has agreed to install or commission the goods, an additional charge consisting of normal Company labor and travel rates may be made by the Company if the Company is prevented from carrying out such installation or commissioning for more than 30 days after delivery of the goods unless due to the act or omission of the Company. Where the Company has agreed to install or commission the goods and is prevented from carrying out such installation or commissioning for more than 60 days after delivery of the goods (unless due to the act or omission of the Company), all warranty periods will begin 60 days after the delivery date.
b) The Customer warrants that it will at all times provide a safe working environment for the Company's employees, agents or sub-contractors and will comply with all statutory or other regulations in connection therewith and will indemnify the Company in respect to any loss incurred by the Company due to the Customer's breach of this warranty.
c) The Company reserves the right to sub-contract the installation of the goods or the performance of any other services required under the Contract.
d) All goods remaining the property of the Company and delivered on site shall be the responsibility of the Customer until the Company has removed the same. The Customer warrants that it has sufficient insurance coverage to comply with the provision of this sub-clause.

TRADE MARK NAMES
The Customer will be fully responsible for any consequences of any patent, trademark, service mark, design or copyright infringement of the third party right resulting from the Customer's use or sale of any of the goods where they have been modified by the Customer and the Customer shall fully and completely indemnify the Company for all damages, costs, charges and expenses as a result of any such infringement or alleged infringement.

NEGLIGENCE
The Company shall not be liable to the Customer for any damage or for any direct or consequential loss incurred by the Customer in consequence of any negligence on the part of the Company or negligence or willful default on the part of its servants or agents or subcontractors in or in connection with the supply of any goods or in the preparation or provision of any information or advice.

INSURANCE
The Company will maintain comprehensive general liability, automotive liability and property damage coverage for injuries to persons and property occurring because of the performance of services by Company under this Contract, with the limits of $6,000,000 for injuries and $1,000,000 for damage to property. Such insurance shall constitute the limits of Company liability to Customer for such injuries and damage.

GOVERNING LAW
This Contact shall be governed by and construed in accordance with the laws of Maryland. Any action stemming or arising from this Contract shall be brought exclusively to a court of competent jurisdiction in Maryland. Customer waives the right to change of venue and further consents to the exercise of personal jurisdiction by any such Maryland court with respect to any such proceeding. Except in cases of non-payment, neither party may institute any action in any form arising out of these Terms and Conditions more than one (1) year after the cause of action has arisen.